Experiment No. 1

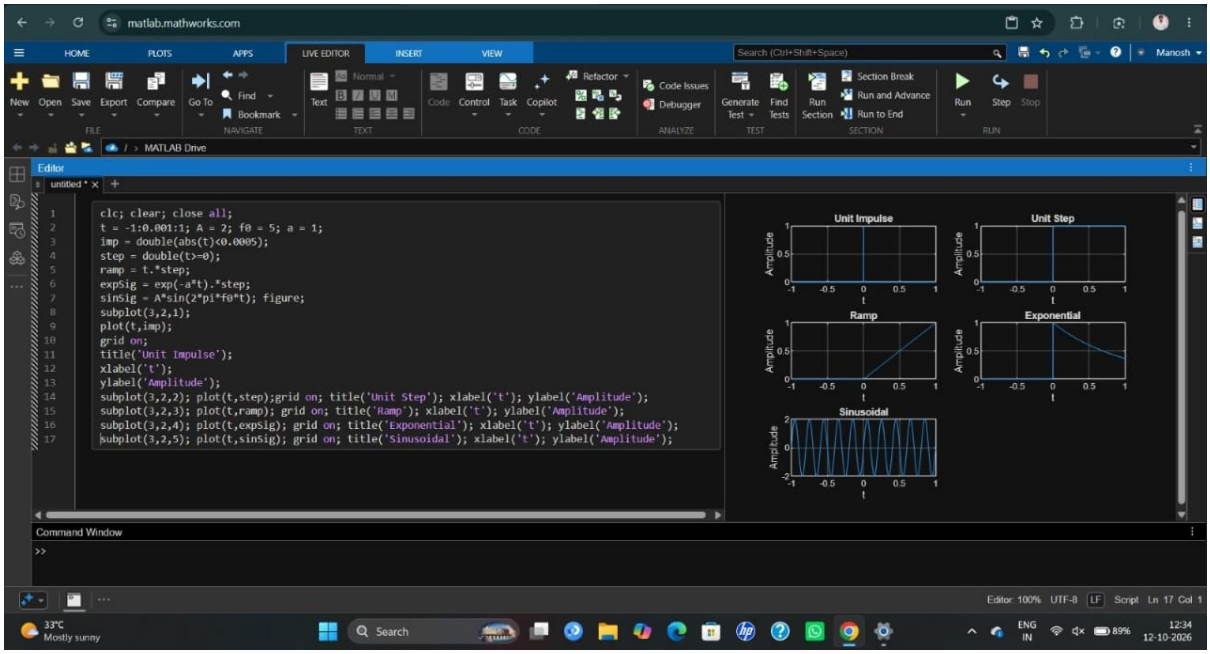
Generation and Classification of Continuous-Time and Discrete-Time Signals

(i) To generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal, and exponential signals.

MATLAB Coding

% Experiment 1: Generation of Continuous-Time Signals

```
clc; clear; close all;
t = -1:0.001:1; A = 2; f0 = 5; a = 1;
imp = double(abs(t)<0.0005);
step = double(t>=0);
 ramp = t.*step;
 exp Sig = exp(-a*t).*step;
sin Sig = A*sin(2*pi*f0*t);
 figure;
subplot(3,2,1); plot(t,imp); grid on; title('Unit Impulse'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,2); plot(t,step); grid on; title('Unit Step'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,3); plot(t,ramp); grid on; title('Ramp'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,4); plot(t,expSig); grid on; title('Exponential'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,5); plot(t,sinSig); grid on; title('Sinusoidal'); xlabel('t'); ylabel('Amplitude');
```



(ii) To generate and visualize basic discrete-time signals and analyze their amplitude and time-index characteristics.

MATLAB Coding

```
% Experiment 2: Generation of Discrete-Time Signals

clc; clear; close all;
n = -20:20;
A = 2; f0 = 0.1;
a = 0.1; imp = double(n==0);
step = double(n>=0);
ramp = n.*step;
expSig = (0.8).^n .* step;
sinSig = A*sin(2*pi*f0*n);
figure;
subplot(3,2,1); stem(n,imp,'filled'); grid on; title('Unit Impulse');
subplot(3,2,2); stem(n,step,'filled'); grid on; title('Unit Step');
subplot(3,2,3); stem(n,ramp,'filled'); grid on; title('Ramp');
subplot(3,2,4); stem(n,expSig,'filled'); grid on; title('Exponential');
subplot(3,2,5); stem(n,sinSig,'filled'); grid on; title('Discrete Sinusoid'); xlabel('n');
```

